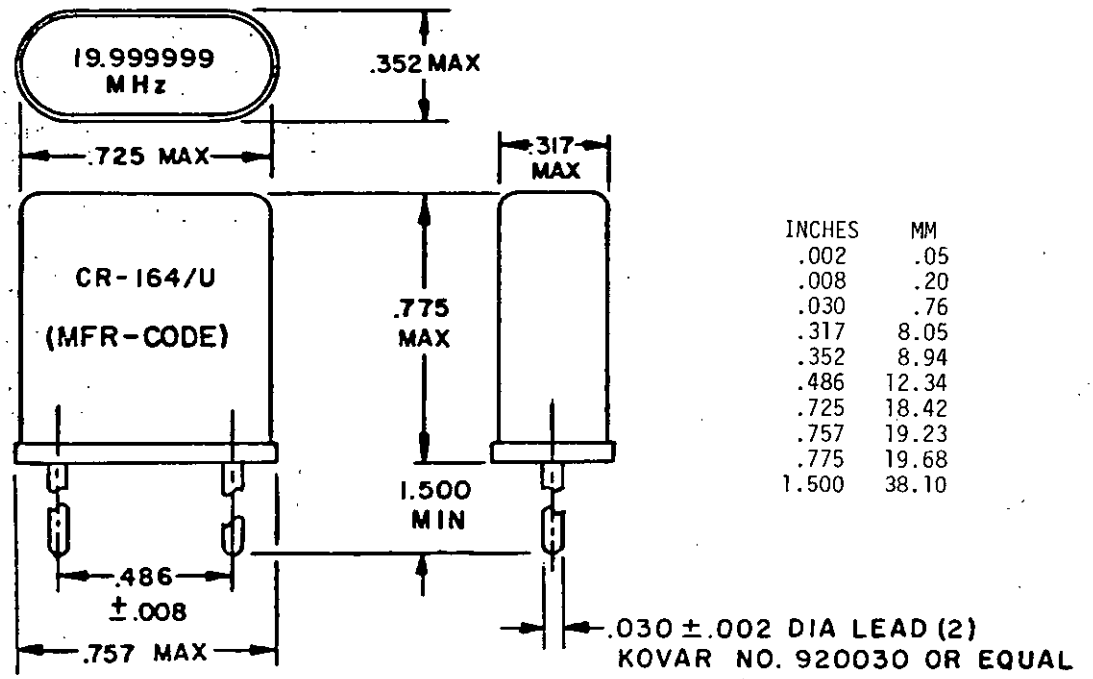
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-164/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; series.



INCHES	MM
.002	.05
.008	.20
.030	.76
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.68
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The wire leads shall be tinned, suitable for soldering.

FIGURE 1. Dimensions and configuration.

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FSC 5955

TABLE I. Requirements.

Holder	HC-33/U (MIL-H-10056/20)	
Dimensions, marking, and configuration	See figure	
Frequency range	0.8 to 20 MHz, inclusive	
Frequency tolerance	±50 PPM	
Equivalent resistance	See table III	
Mode of oscillation	Fundamental	
Operating temperature range	-55° to +105°C, inclusive	
Test set, calibration values, and rated drive level	See table II	
Capacitance, shunt	7 pF, maximum	
Shock (specified pulse):	0.8 to 2.0 MHz	2.0+ to 20 MHz
Frequency change permitted	±10 PPM	±5 PPM
Equivalent-resistance change permitted	±15 percent	±10 percent
Vibration: 1/	0.8 to 2.0 MHz	2.0+ to 20 MHz
Frequency change permitted	±10 PPM	±5 PPM
Equivalent-resistance change permitted	±15 percent	±10 percent
Thermal shock:	0.8 to 2.0 MHz	2.0+ to 20 MHz
Frequency change permitted	±10 PPM	±5 PPM
Equivalent-resistance change permitted	±15 percent	±10 percent
Aging:		
Frequency change permitted	±5 PPM	

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	0.8 to 1.5	100	10	---	
	1.5+ to 2.25	50	15	---	
	2.25+ to 3.4	40	15	---	10.0 ±2.0
	3.4+ to 5.1	25	20	---	
	5.1+ to 7.5	14	25	---	
	7.5+ to 10	11	30	---	
	10+ to 15	13	20	---	5.0 ±1.0
TS-683/TSM	15+ to 20	10	--	0.22	5.0 ±1.0

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TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance	Frequency range, inclusive	Maximum resistance
MHz	Ohms	MHz	Ohms
0.8 to 0.85	630	2.12+ to 2.25	250
0.85+ to 0.9	600	2.25+ to 2.6	200
0.9+ to 1	580	2.6+ to 3	150
1+ to 1.12	540	3+ to 3.4	110
1.12+ to 1.25	490	3.4+ to 3.75	90
1.25+ to 1.37	450	3.75+ to 4	75
1.37+ to 1.5	410	4+ to 5	60
1.5+ to 1.62	380	5+ to 7	35
1.62+ to 1.75	330	7+ to 10	24
1.75+ to 1.87	300	10+ to 15	22
1.87+ to 2	290	15+ to 20	20
2+ to 2.12	270		

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 19
DLA - ES

User activities:
Navy - AS, MC
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0606-5)

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